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Research paper

Prediction of the stress distributions and strength properties of 3-D braided composites with an eccentric circular hole

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ABSTRACT

Due to their low delaminating tendency and high damage tolerance, 3-D braided composites are capable of being utilized in notched components. However, a few studies have investigated the mechanical properties of braided composites with open holes in multi-scale. In this paper, a multi-scale study on the mechanical properties of 3-D 4-directional braided composites with an eccentric circular hole is carried out. The equivalent elastic properties of 3-D 4-directional braided composites are calculated by the homogenization method based on the geometry of the meso unit cells. According to Lekhnitskii's theory, the local stress correction (LSC) method for computing the transverse stress around the eccentric hole is proposed. The models of meso unit cells with different braided angles are established, and the strength of the defect-free composites is predicted by the Tsai-Wu criterion. In the macroscopic view, using the point stress criterion (PSC), the average stress criterion (ASC), and numerical analysis, the strength properties of braided composites with different hole radii, braided angles, and eccentric distances are predicted. This study can be utilized to estimate the stress distributions and strength properties of 3-D braided composites with an eccentric hole and provides theoretical and simulation foundations for the problems of orthotropic materials with holes.

1. Introduction

Compared with metal alloy materials, tranditional laminated composites and natural biological materials (Li et al., 2019; Zou et al., 2009), 3-D braided composites represent high shear stiffness, transverse strength, delamination resistance, damage tolerance, and impact resistance (Dow and Dexter, 1997). The braided composites can hold most of their strength and proceed to function even if damaged, making them suitable for structures with holes and slots. Therefore, the research on mechanical properties of 3-D braided composites with defects possesses theoretical and engineering significance.

The stiffness and strength properties of 3-D 4-directional braided composites are analyzed based on the mesostructures theory. In the early stage, the fiber interlock model (Ma et al., 1986), fiber inclination model (Yang et al., 1986), and fiber helix model (Kalidindi and Franco, 1997) were built. With the development of CAD technology, some meso unit cell models have been proposed considering the morphology, interaction, and interface of constituent materials (Chen et al., 1999; Pandey and Hahn, 1996). The equivalent stiffness of 3-D braided composites can be obtained based on mesostructure studies. The classical stiffness prediction methods for braided composites include the strain energy method (Ma et al., 1986), the laminated plate theory (Yang et al., 1986), the three-cell model (Wu, 1996), the weighted-average model method (Kalidindi and Franco, 1997), the finite multi-phase element method (Chen et al., 1999), and homogenization theory (Feng et al., 2001). Recently, new models for calculating the equivalent elastic properties of braided composites have been developed. In 2015, Zhang et al. (2015a) proposed the fiber-embedded matrix method and computed the equivalent stiffness of 3-D braided composites from a new point of view. Taking into account the inherent defects, Xu et al. (Xu and Qian, 2014) and Dong et al. (Dong and Huo, 2016) designed different representative unit cell models of 3-D braided composites, respectively.

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Zhang et al., 2017a, 2017b discussed the influence of the interface between component constituents and attained the stiffness properties of braided composites by adopting periodic mesh and free mesh in Abaqus.

For laminated composites, two categories of strength criteria are generally used to diagnose their initial damage. The first group are the phenomenological methods that do not distinguish failure modes (Tsai and Wu, 1974; Azzi and Tsai, 1965; Wang et al., 2016; Hoffman, 1967; Jiang et al., 2013), mainly including the Tsai-Hill, Tsai-Wu, and Hoffman criteria. The second category of failure criteria is based on physical mechanisms, which can describe different failure modes (Puck and Schürmann, 1998; Hashin, 1980; Zhang et al., 2015b, 2015c), including the Hashin and Puck criteria. Dedicated strength criteria for 3-D braided composites have not been constructed so far, and most of the existing works are based on the theories developed for laminated composites (Zeng et al., 2004a; Gu, 2004; Li et al., 2008). Fang et al. (2009) put forward the nonlinear constitutive damage theory of 3-D 4-directional braided composites and pointed out that the damage evolvement led to the ultimate failure of the composites. Adopting homogenization theory, Dong et al. (Dong and Feng, 2010) took into account the interior-, surface- and corner cells and simulated the damage evolution of 3-D braided composites. Based on the 3D digital image correlation (DIC), a methodology for a novel assessment of the fracture mechanics of SiC/SiC ceramic matrix composites was presented (Bumgardner et al., 2019, 2021; Heim et al., 2020), and targeted experiments on composites in different environments were carried out. Zeng et al. (2004b) utilized the multi-phase finite element method to predict the strength properties of braided composites according to the nonlinear stress-strain relationship.

Currently, most studies on anisotropic plates with holes focus primarily on laminated composites, and a complete theoretical system for 3-D braided composites has not been established. In 1968, Lekhnitskii (1968) analyzed the stress distributions of infinite anisotropic plates with a hole bearing different loads and laid the theoretical foundation for such problems. In the 1970s, Whitney and Nuismer et al. (Konish and Whitney, 1975; Whitney and Nuismer, 1974; Nuismer, 1975) studied the strength properties of fiber-reinforced composites with a circular hole and proposed the well-known point stress criterion (PSC) and average stress criterion (ASC), respectively. On the basis of the PSC and ASC methods, distinct strength criteria for laminated plates with holes are put forward (Tan, 1988; Jk K Takeda and Ds, 1995; Srivastava and Kumar, 2002). Crew et al. (Crews and Raju, 1981) discussed the state of stress around the hole in laminated plates and explored the stress concentration of orthotropic materials. Chang et al. (Chang and Chang, 1987; Chang et al., 1991) established and improved the progressive damage model of notched laminates, which can be used to predict the failure mode of composites under tension and compression.

Studies on braided composites with defects have been gradually carried out recently. Tate and Hwan et al. (Tate et al., 2006; Hwan et al., 2011) calculated the strength of 2-D braided composites with holes by revising PSC and ASC methods. In 2017, Saleh (Saleh et al., 2017) obtained the mechanical properties of woven composites under eccentric load based on tensile experiments. Harrell et al. (Love-Baker et al., 2021; Harrell et al., 2022) dissussed the influence of the off-axis angles and porosities of the fiber properties, and an analytical relationship between fiber misalignment angle and measured properties was developed. According to the tensile and compression tests, Liang et al., 2020, 2021 analyzed the macro mechanical properties of 3-D 4-directional and 5-directional braided composites with a central circular hole on a macro scale and measured the strength of composites with different hole radii.

In this study, based on the geometric configuration of three types of meso unit cells, the macro equivalent elastic properties of 3-D 4-directional braided composites are obtained by the homogenization theory. The meso unit cell models of braided composites with different braided angles are established by adopting the octagonal cross-section assumption. Tsai-Wu criterion is then utilized to diagnose the initial damage, and the strength properties of defect-free 3-D braided composites are simulated in Abaqus. According to Lekhnitskii's theory, the local stress correction (LSC) method for calculating the transverse stress of orthotropic materials with an eccentric hole is proposed. The strength of 3-D 4-directional braided composites with a hole is obtained by different failure criteria and compared with the existing experimental results. Furthermore, the strength properties of braided composites with an eccentric hole are predicted, and the influences of braided angles, hole radii, and eccentric distances are discussed.

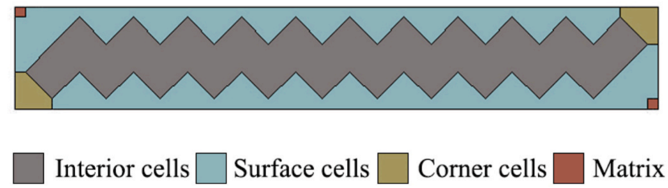

Fig. 1. Areas of three types of unit cells.

2. Equivalent elastic properties of 3-D braided composites

2.1. Unit cells of braided composites

In this paper, periodic unit cell models are used to describe the mechanical properties of 3-D 4-directional braided composites. The meso unit cells in braided composites mainly include interior-, surface-, and corner cells. The three types of unit cells occupy different positions and volume fractions in the composites, as shown in Fig. 1.

The cross-section of the braided composites is a rectangular $m \times n$ structure. For the cases that m and n are both even numbers or both or one of them are odd numbers, the volume fraction of the three types of unit cells can be calculated by Equations (1) and (2), respectively.

$$V_i = \frac{2(\mathrm{m}-1)(\mathrm{n}-1)+2}{2mn+m+n}, V_s = \frac{3(\mathrm{m}+\mathrm{n}-4)}{2mn+m+n}, V_c = \frac{8}{2mn+m+n} \tag{1}$$

$$V_i = \frac{2(\mathrm{m}-1)(\mathrm{n}-1)}{2mn+m+n}, V_s = \frac{3(\mathrm{m}+\mathrm{n}-2)}{2mn+m+n}, V_c = \frac{4}{2mn+m+n} \tag{2}$$

in which, V_i, V_s and V_c are the volume fractions of interior-, surface- and corner cells in the braided composites, respectively.

The fiber bundles in the three types of unit cells display different trends, as shown in Fig. 2, in which, γ, θ and β are the interior braided-, surface braided-, and corner braided angles, respectively, and,

$$\pi \tan \gamma = 12 \tan \theta = 12 \tan \beta \tag{3}$$

By analyzing the geometric configurations of three types of unit cells, the contact between the macro and meso levels of 3-D braided composites could be established, and thus a multi-scale model for the mechanical properties of these composites can be developed.

2.2. Computation method of the equivalent elastic properties

The homogenization method is used to calculate the stiffness of the unit cells. Using the weighted summing of the stiffness matrices of the three types of unit cells, the equivalent elastic properties of 3-D 4-directional braided composites can be obtained. Homogenization theory is a mathematical method expressing any structural field as expansions from macro and micro scales. According to the perturbation equations attained from the basic equations and boundary conditions of continuum mechanics, the macro equivalent elastic properties of braided composites can be solved.

The scale of unit cells is much smaller than that of the overall structure, which could be set to ε order, where $0 < \varepsilon \ll 1$. It can be considered that the field variable $\mathbf{\Phi}(x)$ depends on both the macro coordinate x and meso scale coordinate $y = x/\varepsilon$, and (Feng et al., 2001),

$$\mathbf{\Phi}(x) = \mathbf{\Phi}(x, y), y = x / \varepsilon \tag{4}$$

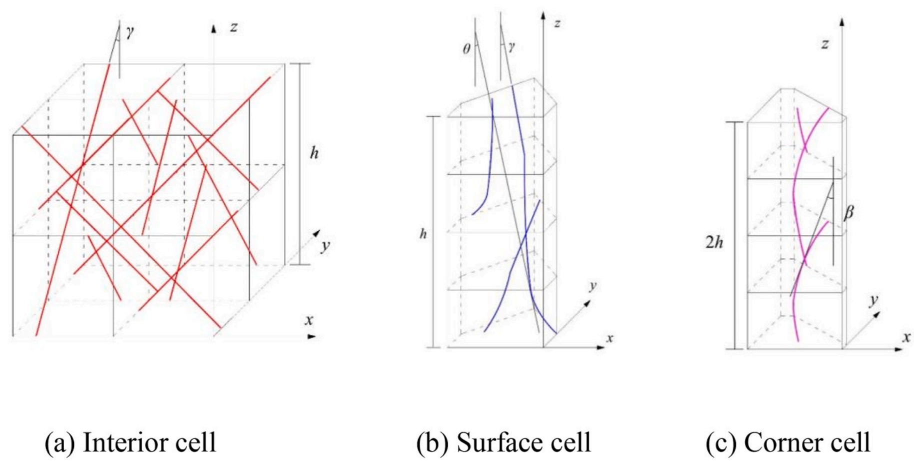

Fig. 2. Directions of fiber bundles in unit cells.

Table 1
Parameters of the constituent materials (Xu et al., 2018).

Material	T700-12K Yarn						TED-86 Epoxy resin	
Material parameter	E_{11} (GPa)	E_{22} (GPa)	G_{12} (GPa)	G_{23} (GPa)	μ_{12}	μ_{23}	E (GPa)	μ
Value	230	130	15	20	0.28	0.4	3.45	0.35

The partial differential equations of structural field functions with respect to coordinates can be expressed as:

$$\frac{\partial \mathbf{\Phi}(x)}{\partial x_i} = \frac{\partial \mathbf{\Phi}(x,y)}{\partial x_i} + \frac{1}{\varepsilon}\frac{\partial \mathbf{\Phi}(x,y)}{\partial y_i} \tag{5}$$

According to the basic equations of elasticity, the strain and stress can be written as (Dong and Feng, 2010):

$$e_{ij}(x) = \varepsilon^{-1} e_{ij}^{-1}(x,y) + e_{ij}^{(0)}(x,y) + \varepsilon e_{ij}^{(1)}(x,y) + \varepsilon^2 e_{ij}^{(2)}(x,y) + \dots \tag{6}$$

$$\sigma_{ij}(x) = \varepsilon^{-1} \sigma_{ij}^{-1}(x,y) + \sigma_{ij}^{(0)}(x,y) + \varepsilon \sigma_{ij}^{(1)}(x,y) + \varepsilon^2 \sigma_{ij}^{(2)}(x,y) + \dots \tag{7}$$

in which, $e_{ij}^{(m)}(x,y)$ and $\sigma_{ij}^{(m)}(x,y)$ are the multiplier of ε^m. Considering the equilibrium equations and the periodicity of the boundary, it can be attained:

$$\left(C_{ijkl}\cdot\left(e_{klx}\left(u^{(0)}\right) + e_{kly}\left(u^{(1)}\right)\right)\right)_{,y_j} = 0 \tag{8}$$

where, u is the displacement field, $e_{klx} = \frac{1}{2}\left(\frac{\partial}{\partial x_k} + \frac{\partial}{\partial x_l}\right)$.

The equivalent displacement, $\chi_i^{kl}(y)$, is utilized to separate macro and meso variables. The equivalent stiffness matrix of the unit cell could be computed by averaging the meso stress field over the volume of the unit cell.

$$C_{ijkl}^R = \frac{1}{|\Omega|}\int_\Omega C_{ijkl} + C_{ijkl}\cdot e_{ijy}\left(\chi^{kl}(y)\right)d\Omega \tag{9}$$

in calculating the elastic constants of the braided composites, the fiber bundles are generally simplified into linear elastic and transversely isotropic material, while the matrix is considered linear elastic, homogeneous and isotropic material. The equivalent stiffness matrices of different unit cells can be obtained according to Equation (9), and the macro equivalent stiffness matrix of the 3-D braided composites is then attained by:

$$[C] = V_i\left[C^i\right] + V_s\left[C^s\right] + V_c\left[C^c\right] \tag{10}$$

where, $[C^i]$, $[C^s]$ and $[C^c]$ are the equivalent stiffness matrix of interior-, surface- and corner cells, respectively.

2.3. Stiffness prediction of the 3-D 4-directional braided composites

Adopting the stiffness volume averaging method, the elastic constants of 3-D 4-directional braided composites are computed in reference (Xu et al., 2018), and the parameters of the constituent materials are shown in Table 1. Table 2 shows the equivalent elastic constants of 3-D 4-directional braided composites obtained according to the homogenization theory and the stiffness volume averaging method, along with the experimental results from (Xu et al., 2018). It could be concluded from

Table 2
Computational results of equivalent elastic constants.

Parameters		E_x/GPa	E_z/GPa	G_{zx}/GPa	G_{xy}/GPa	μ_{zx}	μ_{xy}
$\gamma = 20°$	Experiment (Xu et al., 2018)	8.560	94.750			0.787	
	Stiffness volume average method	14.355	93.327	19.123	6.415	0.510	0.314
	Homogenization method	10.322	93.315	25.581	4.845	0.755	0.344
$\gamma = 30°$	Experimen (Xu et al., 2018)	10.150	43.250			0.703	
	Stiffness volume average method	15.844	46.605	18.551	9.261	0.488	0.281
	Homogenization method	12.563	44.900	15.098	8.771	0.682	0.302
$\gamma = 45°$	Experiment (Xu et al., 2018)	15.410	25.400			0.480	
	Stiffness volume average method	19.522	28.806	15.655	15.214	0.353	0.323
	Homogenization method	16.650	26.706	11.443	13.078	0.414	0.368

Table 3
Parameters of the constituent materials (Liang et al., 2020).

Properties	Matrix/Epoxy resin	Fiber/AS4-6k
Longitudinal tensile modulus (GPa)	2.74	231
Transverse tensile modulus (GPa)	–	15
Longitudinal shear modulus (GPa)	–	24
Transverse shear modulus (GPa)	–	5.03
Longitudinal Poisson's ratio	0.35	0.28
Longitudinal tensile strength (MPa)	80	4850
Shear strength (MPa)	60	–

Table 4
Equivalent elastic constants of 3-D 4-directional braided composites.

Braided angle(°)	E_x(GPa)	E_y(GPa)	G_{xy}(GPa)	G_{yz}(GPa)	μ_{xy}	μ_{yz}
12	91.74	9.44	14.58	4.15	0.28	0.47
20	85.47	11.14	14.99	4.52	0.29	0.37
30	67.57	15.74	16.64	5.98	0.34	0.32
40	42.19	20.33	21.19	9.84	0.35	0.23

the table that, compared with the stiffness volume averaging method, the homogenization theory represents better agreement with the experiments. Thus, homogenization theory is adopted to evaluate the equivalent stiffness and is used for analyzing the mechanical properties of braided composites with an eccentric hole.

The material parameters in reference (Liang et al., 2020) are applied for analyzing the mechanical properties of braided composites with an eccentric hole, as shown in Table 3. Based on the homogenization theory, the macro equivalent stiffness of 3-D 4-directional braided composites with a dimension of 100mm × 25mm × 3 mm at braided angles of 12°, 20°, 30°, and 40° are calculated and shown in Table 4.

3. Transverse stress distributions around an eccentric hole

3.1. State of the transverse stress

The transverse stress distribution around a circular hole of the infinite composite plates under tension is shown in Fig. 3, which can be expressed as (Lekhnitskii, 1968):

$$\sigma_y(x,0)=\frac{\sigma}{2}\left\{2+\left(\frac{R}{x}\right)^2+3\left(\frac{R}{x}\right)^4-\left(K_t^{\infty}-3\right)\cdot\left[5\left(\frac{R}{x}\right)^6-7\left(\frac{R}{x}\right)^8\right]\right\} \tag{11}$$

where, σ is the tensile stress at infinity, R is the radius of the circular hole and K_T^{∞} is the stress concentration factor. For orthotropic materials (Lekhnitskii, 1968),

$$K_t^{\infty}=1+\sqrt{2\left(\sqrt{\frac{E_{11}}{E_{22}}}-\nu_{12}\right)+\frac{E_{22}}{G_{12}}} \tag{12}$$

For the orthotropic plate shown in Fig. 4 with the width of $2b$, the hole radius of R, and the eccentric distance of δ, Equation (11) is revised using coefficient C_A in reference (Wang and Song, 2003). The stress concentration factor of the plate with an eccentric hole at Point A can be obtained:

$$K_{tA}=K_t^{\infty}C_A \tag{13}$$

in which,

$$C_A=\frac{H_4}{H_1H_4+H_2H_3} \tag{14}$$

The terms in Equation (14) are:

$$\begin{aligned}
H_1&=\frac{b-\delta}{2b}-\frac{R^2}{4b(b-\delta)}-\frac{R^4}{4b(b-\delta)^3}+\frac{\left(K_t^{\infty}-3\right)R^6}{4b(b-\delta)^5}-\frac{\left(K_t^{\infty}-3\right)R^8}{4b(b-\delta)^7}\\
H_2&=\frac{b+\delta}{2b}-\frac{R^2}{4b(b+\delta)}-\frac{R^4}{4b(b+\delta)^3}+\frac{\left(K_t^{\infty}-3\right)R^6}{4b(b+\delta)^5}-\frac{\left(K_t^{\infty}-3\right)R^8}{4b(b+\delta)^7}\\
H_3&=b^2-\delta^2+R^2\left(0.5-\frac{K_t^{\infty}-3}{12}+\ln\frac{b-\delta}{R}\right)-\frac{R^2\delta}{b-\delta}-\frac{3R^4}{2(b-\delta)^2}-\frac{R^4\delta}{(b-\delta)^3}\\
&+\frac{5\left(K_t^{\infty}-3\right)R^6}{4b(b-\delta)^4}+\frac{\left(K_t^{\infty}-3\right)R^6\delta}{(b-\delta)^5}-\frac{7\left(K_t^{\infty}-3\right)R^8}{6b(b-\delta)^6}-\frac{\left(K_t^{\infty}-3\right)R^8\delta}{(b-\delta)^7}\\
H_4&=b^2-\delta^2+R^2\left(0.5-\frac{K_t^{\infty}-3}{12}+\ln\frac{b+\delta}{R}\right)+\frac{R^2\delta}{b+\delta}-\frac{3R^4}{2(b+\delta)^2}+\frac{R^4\delta}{(b+\delta)^3}\\
&+\frac{5\left(K_t^{\infty}-3\right)R^6}{4b(b+\delta)^4}-\frac{\left(K_t^{\infty}-3\right)R^6\delta}{(b+\delta)^5}-\frac{7\left(K_t^{\infty}-3\right)R^8}{6b(b+\delta)^6}+\frac{\left(K_t^{\infty}-3\right)R^8\delta}{(b+\delta)^7}
\end{aligned} \tag{15}$$

The theory represents certain irrationality in the calculation because the position of the hole only affects the stress distribution in the neighborhood of the hole, and the transverse stress far away from the hole is approximately equal to σ. Hence it is not necessary to correct the stress in the whole width of the plate, and the more reasonable way is to revise the transverse stress within a certain distance from the hole edge. Assume that the influence distances of stress concentration caused by the eccentricity on AA' and BB' are $2d_A$ and $2d_B$, respectively, according to the forces and moments balance conditions of the plate with an eccentric hole,

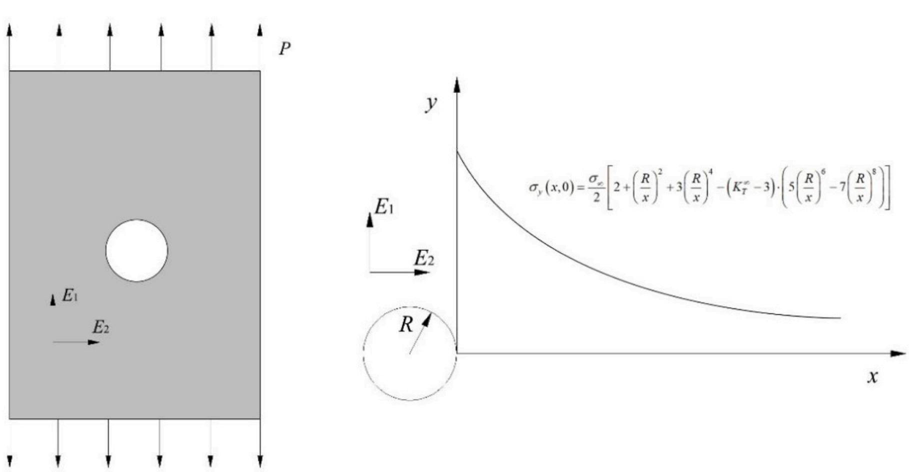

Fig. 3. Transverse stress distribution around the hole of composites.

$$\begin{aligned}&\int_{R}^{R+2d_A} C_A^{'}\frac{\sigma}{2}\left\{2+\left(\frac{R}{x}\right)^2+3\left(\frac{R}{x}\right)^4-\left(K_T^{\infty}-3\right)\cdot\left[5\left(\frac{R}{x}\right)^6-7\left(\frac{R}{x}\right)^8\right]\right\}dx+\int_{R+2d_A}^{b-\delta}\sigma dx+\\&\int_{R}^{R+2d_B} C_B^{'}\frac{\sigma}{2}\left\{2+\left(\frac{R}{x}\right)^2+3\left(\frac{R}{x}\right)^4-\left(K_T^{\infty}-3\right)\cdot\left[5\left(\frac{R}{x}\right)^6-7\left(\frac{R}{x}\right)^8\right]\right\}dx+\int_{R+2d_B}^{b+\delta}\sigma dx=2b\sigma\end{aligned} \tag{16}$$

$$\begin{aligned}&\int_{R}^{R+2d_A} C_A^{'}\frac{\sigma}{2}(x+\delta)\left\{2+\left(\frac{R}{x}\right)^2+3\left(\frac{R}{x}\right)^4-\left(K_T^{\infty}-3\right)\cdot\left[5\left(\frac{R}{x}\right)^6-7\left(\frac{R}{x}\right)^8\right]\right\}dx-\\&\int_{R}^{R+2d_B} C_B^{'}\frac{\sigma}{2}(x-\delta)\left\{2+\left(\frac{R}{x}\right)^2+3\left(\frac{R}{x}\right)^4-\left(K_T^{\infty}-3\right)\cdot\left[5\left(\frac{R}{x}\right)^6-7\left(\frac{R}{x}\right)^8\right]\right\}dx+\\&\int_{R+2d_A}^{b-\delta}\sigma(x+\delta)dx-\int_{R+2d_B}^{b+\delta}\sigma(x-\delta)dx=0\end{aligned} \tag{17}$$

where, $C_A^{'}$ and $C_B^{'}$ are the correction coefficients on AA' and BB', respectively. Solving Equations (16) and (17), it can be attained:

$$C_A^{'}=\frac{G_2+G_4}{G_1G_4+G_2G_3},\ C_B^{'}=\frac{G_3-G_1}{G_1G_4+G_2G_3} \tag{18}$$

in which,

$$\begin{aligned}G_1&=\frac{2R+4d_A-\frac{R^2}{R+2d_A}-\frac{R^4}{(R+2d_A)^3}+\left(K_t^{\infty}-3\right)\left[\frac{R^6}{(R+2d_A)^5}-\frac{R^8}{(R+2d_A)^7}\right]}{4(R+d_A+d_B)}\\G_2&=\frac{2R+4d_B-\frac{R^2}{R+2d_B}-\frac{R^4}{(R+2d_B)^3}+\left(K_t^{\infty}-3\right)\left[\frac{R^6}{(R+2d_B)^5}-\frac{R^8}{(R+2d_B)^7}\right]}{4(R+d_A+d_B)}\\G_3&=\frac{\frac{3}{2}R^2+4Rd_A+4{d_A}^2+R^2\ln\frac{R+2d_A}{R}-\frac{3R^4}{2(R+2d_A)}+\left(K_t^{\infty}-3\right)\left[\frac{5R^6}{4(R+2d_A)^4}-\frac{7R^8}{6(R+2d_A)^6}-\frac{R^2}{12}\right]}{4\left(Rd_A+{d_A}^2+\delta d_A-Rd_B-{d_B}^2+\delta d_B+\delta R\right)}+\\&\frac{\delta\left\{2R+4d_A-\frac{R^2}{R+2d_A}-\frac{R^4}{(R+2d_A)^3}+\left(K_t^{\infty}-3\right)\left[\frac{R^6}{(R+2d_A)^5}-\frac{R^8}{(R+2d_A)^7}\right]\right\}}{4\left(Rd_A+{d_A}^2+\delta d_A-Rd_B-{d_B}^2+\delta d_B+\delta R\right)}\\G_4&=\frac{\frac{3}{2}R^2+4Rd_B+4{d_B}^2+R^2\ln\frac{R+2d_B}{R}-\frac{3R^4}{2(R+2d_B)}+\left(K_t^{\infty}-3\right)\left[\frac{5R^6}{4(R+2d_B)^4}-\frac{7R^8}{6(R+2d_B)^6}-\frac{R^2}{12}\right]}{4\left(Rd_A+{d_A}^2+\delta d_A-Rd_B-{d_B}^2+\delta d_B+\delta R\right)}-\\&\frac{\delta\left\{2R+4d_B-\frac{R^2}{R+2d_B}-\frac{R^4}{(R+2d_B)^3}+\left(K_t^{\infty}-3\right)\left[\frac{R^6}{(R+2d_B)^5}-\frac{R^8}{(R+2d_B)^7}\right]\right\}}{4\left(Rd_A+{d_A}^2+\delta d_A-Rd_B-{d_B}^2+\delta d_B+\delta R\right)}\end{aligned} \tag{19}$$

The stress concentration factor of the orthotropic plate with an eccentric hole could be calculated based on the equations above, and the distribution of the transverse stress σ_y is obtained. The reader can refer to reference (Whitney and Nuismer, 1974) for the values of d_A and d_B while the parameters such as hole radius and the distance of AA' and BB' are case-depended.

3.2. Prediction of the transverse stress

Owing to the difficulty of measuring the stress at any point of the 3-D 4-directional braided composites with an eccentric hole experimentally, the transverse stress attained by the theoretical method could be compared with the numerical simulations in Abaqus. The computing model is constructed on the macro scale due to the integrated structure of the 3-D braided composites. The material parameters in reference (Liang et al., 2020) are utilized in the calculation and simulation, and the equivalent stiffness values of the braided composites are shown in Table 4.

In the simulations, the plate size is 100mm × 25mm × 3 mm. The tensile load along the principal directional of the braided composites is 10 MPa, and the braiding direction is the same as the direction of the tensile load. The parameters of different regions are assigned according to the calculated equivalent elastic constants. The C3D8R solid elements are adopted, and the size of the elements is taken from 0.03 to 0.1 based on the hole radius. One end of the plate model is fixed, and the other end is subjected to uniformly distributed tensile stress, and the transverse stress nephogram of the composite plate with an eccentric hole is obtained.

Fig. 5 shows the state of the transverse stress calculated in Abaqus. It could be seen that the maximum transverse stress in composite plates occurs at the point on the hole edge, which is closest to the plate edge. Tables 5–8 represent the maximum transverse stress in the 3-D 4-directional braided composite plate with an eccentric hole under tension

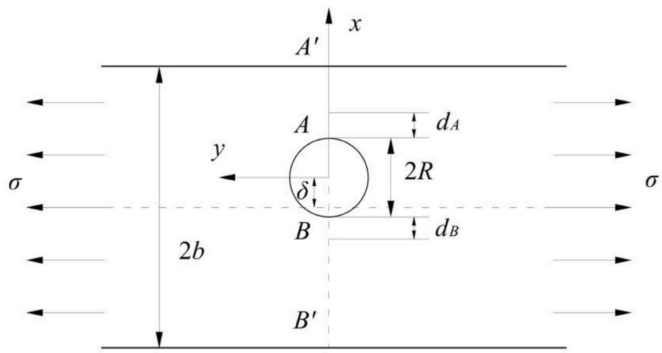

Fig. 4. Orthotropic plate with an eccentric hole under tension.

reported in reference (Wang and Song, 2003), calculated using the proposed LSC method and numerical simulations in this study. It can be concluded that the method proposed in reference (Wang and Song, 2003) is in good agreement with the simulations as the eccentric distance δ is smaller than 3 mm. However, a significant discrepancy shows up when δ/b is greater than 0.3. The stresses computed by the LSC method established in this paper agree with the simulations with different braided angles, hole radii, and eccentric distances. The difference between the two methods can be attributed to the corresponding influence range of stress concentration in the equations for various geometric parameters. In addition, the maximum transverse stress of the 3-D braided composite plates with an eccentric hole increases with the increase of hole radius and eccentric distance and decreases with the increase of braided angle. For composite plates with δ/b smaller than 0.05, the maximum transverse stress is less affected by δ. This phenomenon is due to the stress distribution of the composite plates with a small hole radius being similar to that of infinite plates; therefore, different eccentric distances can not cause significant effects on stress concentration.

Fig. 6 shows the transverse stress curves around the hole in a 3-D 4-directional braided composite with different braided angles and a hole radius of 2.5 mm. It could be concluded that the maximum transverse stress appears at the hole edge and gradually decreases approximately equal to the tensile stress σ with the increase of x. The closer the hole is to the edge of the plate, the more pronounced the variation of the transverse stress around the hole. When δ/b is in the range of 0~0.2, the transverse stress at the hole edge changes inapparent. When δ/b exceeds 0.6, the maximum stress at the hole edge may reach 5–6 times the average (far field) tensile stress. Therefore, for a braided composite part with high strength requirements, δ/b values less than 0.2 would be preferred.

4. Prediction of notched strength properties

3-D 4-directional braided composites possess complex mesostructure, in which the fiber bundles interlace. The braided composite components show complicated failure modes even under a single load. Various damage modes often exist simultaneously and interact mutually in constituent materials, and it is hard to observe the internal damage process in a mechanical experiment. For 3-D braided composites, finite element simulation is the accurate and effective method for detecting damage and predicting strength. By degrading the stiffness of the component materials, the failure process can be simulated, and the strength properties of the composites can be obtained.

4.1. Strength prediction of braided composites based on the unit cells

Considering the sophisticated mesostructure of the 3-D 4-directional braided composites, the section mainly bearing the loads is the interior cells. Thus, the surface and corner cells are ignored in analyzing the strength properties. The simulated strength of the interior cells is adopted as the basis for strength properties analysis of 3-D 4-directional braided composites with an eccentric hole. In the unit cell models, the cross-section of the fiber bundles is assumed as an octagon, which could describe the interaction between fiber bundles and the matrix. Fig. 7 shows the fiber bundles and matrix in a unit cell of 3-D 4-directional braided composites, and Fig. 8 illustrates the interior cell models with different braided angles.

Hashin criterion and Tsai-Wu criterion are applied to diagnose the initial damage of the fiber bundles in the unit cells. In contrast, the von Mises criterion is adopted as the failure criterion for the matrix. Hashin criterion can be written as

$$\begin{aligned}
&\left(\frac{\sigma_1}{X_{\mathrm{T}}}\right)^2+\left(\frac{\sigma_{12}}{S_{12}}\right)^2+\left(\frac{\sigma_{13}}{S_{13}}\right)^2\geq 1,\left(\frac{\sigma_1}{X_{\mathrm{C}}}\right)^2\geq 1\\
&\left(\frac{\sigma_2+\sigma_3}{Y_{\mathrm{T}}}\right)^2+\left(\frac{1}{S_{23}^2}\right)^2\left(\sigma_{23}^2-\sigma_2\sigma_3\right)+\left(\frac{\sigma_{12}}{S_{12}}\right)^2+\left(\frac{\sigma_{13}}{S_{13}}\right)^2\geq 1\\
&\left(\frac{\sigma_2+\sigma_3}{2S_{12}}\right)^2+\left(\frac{\sigma_2+\sigma_3}{Y_{\mathrm{C}}}\right)\left[\left(\frac{Y_{\mathrm{C}}}{2S_{12}}\right)^2-1\right]+\frac{1}{S_{23}^2}\left(\sigma_{23}^2-\sigma_2\sigma_3\right)+\left(\frac{\sigma_{12}}{S_{12}}\right)^2+\left(\frac{\sigma_{13}}{S_{13}}\right)^2\geq 1
\end{aligned} \tag{20}$$

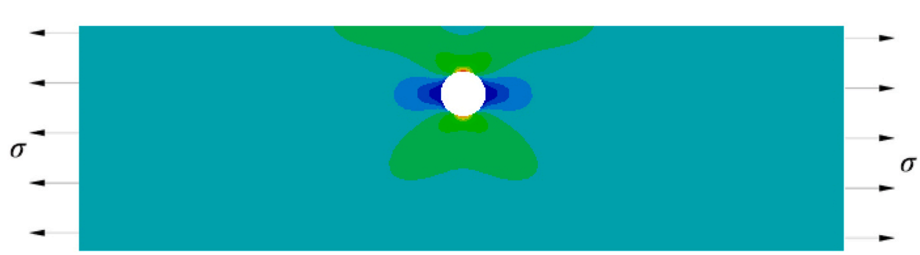

Fig. 5. Stress distribution of 3-D 4-directional braided composites with an eccentric hole.

Table 5
The maximum transverse stress at the hole edge of the plate with a braided angle of 12°.

δ(mm)		0	1.25	2.5	3.75	5	6.25	7.5
$R = 0.5$ mm	Reference (Wang and Song, 2003)	35.20	39.13	44.04	50.36	58.81	70.67	88.56
	LSC method	37.45	37.99	38.67	39.56	40.78	42.56	45.37
	Abaqus	42.12	43.43	43.85	44.12	44.29	44.41	44.68
$R = 1.4$ mm	Reference (Wang and Song, 2003)	35.88	40.00	45.18	51.91	61.01	74.08	94.59
	LSC method	37.29	37.92	38.75	39.90	41.56	44.13	48.56
	Abaqus	43.31	43.78	44.30	44.65	46.52	47.27	47.59
$R = 2.5$ mm	Reference (Wang and Song, 2003)	37.61	42.25	48.15	56.00	67.02	83.92	113.98
	LSC method	38.21	38.94	39.98	41.49	43.85	47.91	56.20
	Abaqus	45.07	46.23	45.61	46.61	48.24	49.80	54.99
$R = 3.9$ mm	Reference (Wang and Song, 2003)	41.80	47.74	55.56	66.55	83.50	114.36	198.58
	LSC method	41.17	42.20	43.75	46.24	50.66	60.03	90.97
	Abaqus	48.55	49.01	49.49	53.66	54.89	63.30	87.32
$R = 4.8$ mm	Reference (Wang and Song, 2003)	46.17	53.56	63.61	78.49	103.74	161.60	–
	LSC method	45.27	46.95	49.58	54.06	62.91	87.31	–
	Abaqus	51.82	52.50	54.11	63.37	68.19	87.22	–

Table 6
The maximum transverse stress at the hole edge of the plate with a braided angle of 20°.

δ(mm)		0	1.25	2.5	3.75	5	6.25	7.5
$R = 0.5$ mm	Reference (Wang and Song, 2003)	34.00	37.39	42.53	48.64	56.80	68.26	85.54
	LSC method	36.17	36.69	37.35	38.21	39.40	41.13	43.87
	Abaqus	39.77	40.18	40.72	41.09	41.60	42.25	43.00
$R = 1.4$ mm	Reference (Wang and Song, 2003)	34.66	38.64	43.64	50.13	58.93	71.55	91.36
	LSC method	36.01	36.62	37.43	38.55	40.16	42.67	47.01
	Abaqus	41.76	43.21	43.25	43.40	44.83	45.51	45.88
$R = 2.5$ mm	Reference (Wang and Song, 2003)	36.35	40.83	46.53	54.12	64.78	81.12	110.25
	LSC method	36.92	37.64	38.64	40.11	42.41	46.40	54.56
	Abaqus	43.32	45.31	44.06	45.02	46.60	48.04	53.29
$R = 3.9$ mm	Reference (Wang and Song, 2003)	40.44	46.19	53.76	64.41	80.87	110.99	194.17
	LSC method	39.81	40.82	42.33	44.76	49.10	58.36	89.22
	Abaqus	45.52	46.10	47.51	51.79	53.40	62.26	84.99
$R = 4.8$ mm	Reference (Wang and Song, 2003)	44.70	51.87	61.62	76.10	100.82	158.05	–
	LSC method	43.81	45.45	48.03	52.43	61.19	85.51	–
	Abaqus	49.77	50.70	52.43	61.06	66.89	85.45	–

Table 7
The maximum transverse stress at the hole edge of the plate with a braided angle of 30°.

δ(mm)		0	1.25	2.5	3.75	5	6.25	7.5
$R = 0.5$ mm	Reference (Wang and Song, 2003)	31.19	34.67	39.02	44.62	52.11	62.62	78.47
	LSC method	33.19	33.67	34.27	35.07	36.17	37.78	40.36
	Abaqus	36.81	37.10	37.53	37.88	38.00	38.34	38.59
$R = 1.4$ mm	Reference (Wang and Song, 2003)	31.80	35.45	40.04	45.99	54.06	65.64	83.82
	LSC method	33.04	33.61	34.35	35.39	36.89	39.24	43.34
	Abaqus	38.07	38.72	38.90	39.21	40.03	40.63	40.89
$R = 2.5$ mm	Reference (Wang and Song, 2003)	33.40	37.51	42.75	49.72	59.53	74.57	101.51
	LSC method	33.92	34.58	35.51	36.89	39.05	42.82	50.68
	Abaqus	39.43	40.32	40.50	40.97	42.35	43.68	48.89
$R = 3.9$ mm	Reference (Wang and Song, 2003)	37.24	42.54	49.53	59.38	74.68	103.00	183.43
	LSC method	36.65	57.58	39.00	41.29	45.42	54.38	84.88
	Abaqus	41.70	42.21	43.77	47.05	49.70	57.09	81.43
$R = 4.8$ mm	Reference (Wang and Song, 2003)	41.24	47.88	56.94	70.47	93.87	149.39	–
	LSC method	40.39	41.93	44.37	48.57	57.06	81.08	–
	Abaqus	45.10	46.33	48.35	55.62	62.16	80.84	–

Table 8
The maximum transverse stress at the hole edge of the plate with a braided angle of 40°.

δ(mm)		0	1.25	2.5	3.75	5	6.25	7.5
$R = 0.5$ mm	Reference (Wang and Song, 2003)	27.78	30.88	34.75	39.75	46.41	55.77	69.89
	LSC method	29.56	29.99	30.54	31.25	32.24	33.71	36.07
	Abaqus	31.40	31.91	32.02	32.40	32.71	33.23	33.69
$R = 1.4$ mm	Reference (Wang and Song, 2003)	28.32	31.57	35.66	40.97	48.15	58.47	74.67
	LSC method	29.44	29.94	30.61	31.54	32.90	35.05	38.85
	Abaqus	32.48	33.00	33.37	33.71	34.05	35.67	36.88
$R = 2.5$ mm	Reference (Wang and Song, 2003)	29.80	33.47	38.15	44.37	53.13	66.58	90.80
	LSC method	30.26	30.86	31.70	32.95	34.93	38.42	45.83
	Abaqus	33.21	33.30	33.87	34.76	35.82	37.51	43.09
$R = 3.9$ mm	Reference (Wang and Song, 2003)	33.33	38.07	44.34	53.20	67.05	93.05	169.49
	LSC method	32.77	33.62	34.91	37.02	40.86	49.36	79.13
	Abaqus	35.57	36.12	37.80	40.44	44.13	52.28	76.19
$R = 4.8$ mm	Reference (Wang and Song, 2003)	36.99	42.97	51.16	63.48	85.13	138.13	–
	LSC method	36.19	37.60	39.85	43.78	51.85	75.25	–
	Abaqus	38.07	39.62	42.22	46.93	55.50	75.88	–

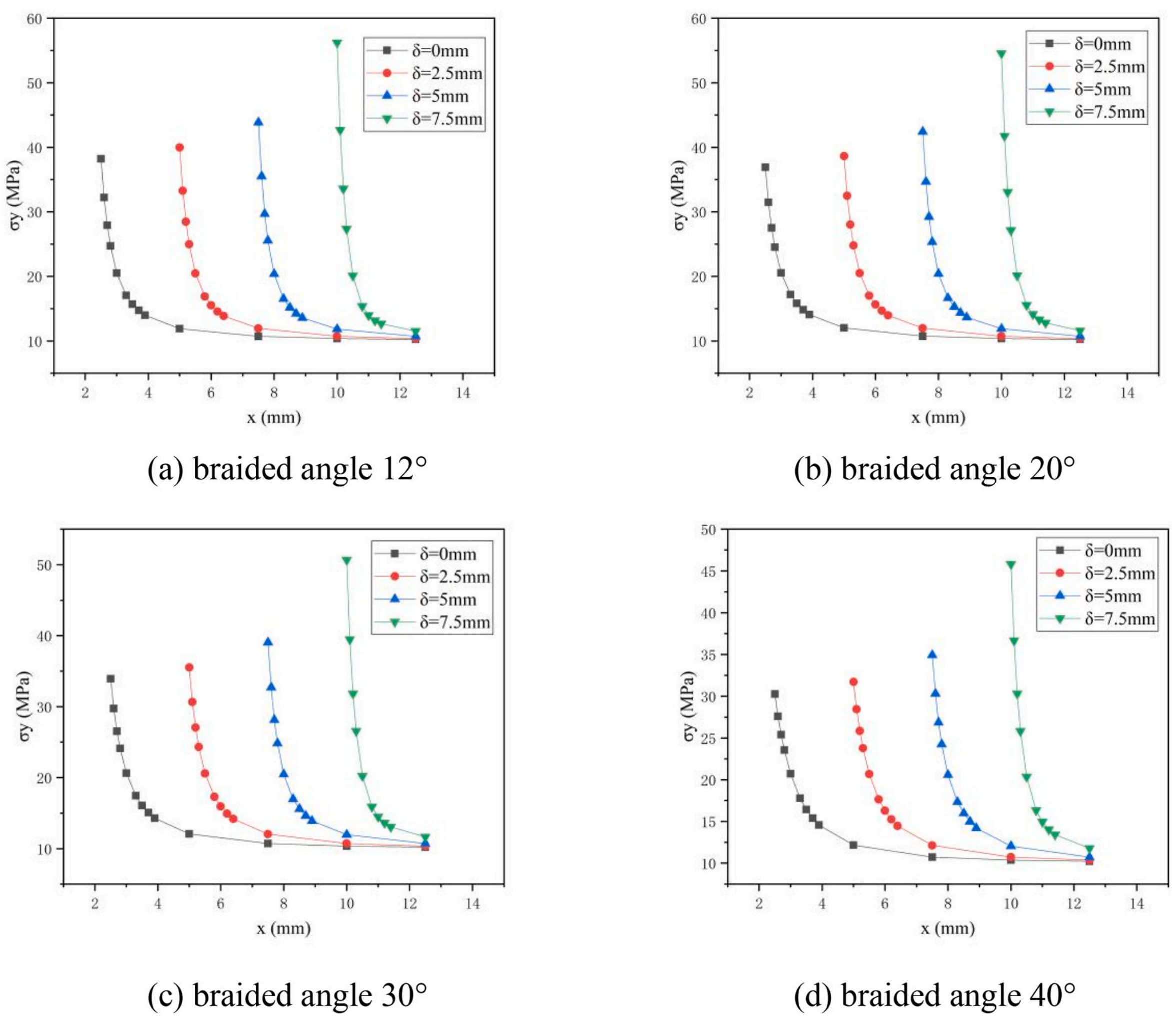

Fig. 6. Transverse stress around the hole in braided composites with a hole radius of 2.5 mm.

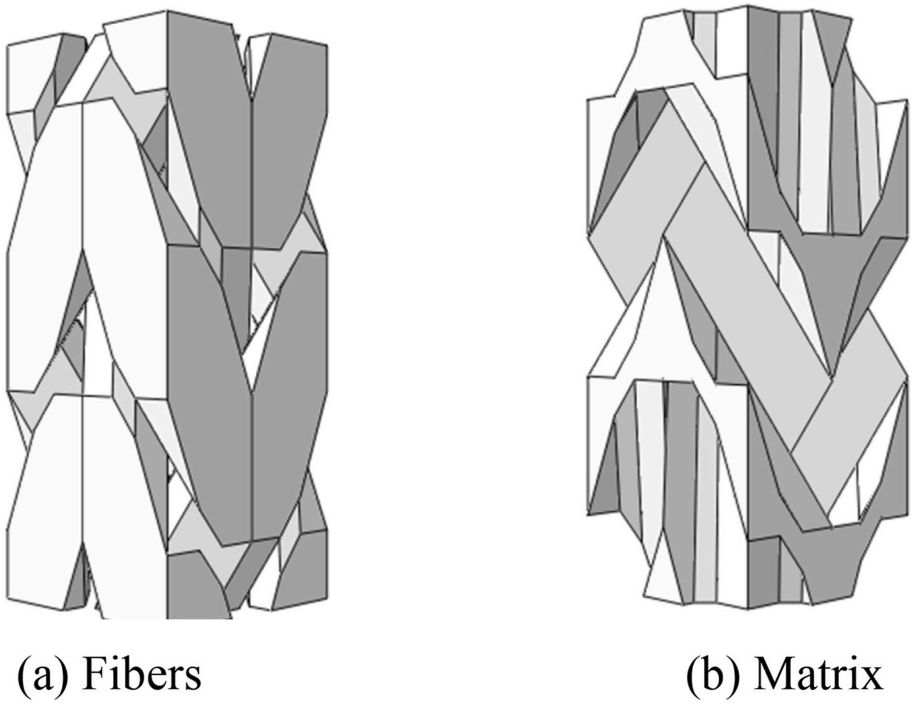

Fig. 7. Component material models in an interior cell of braided composites.

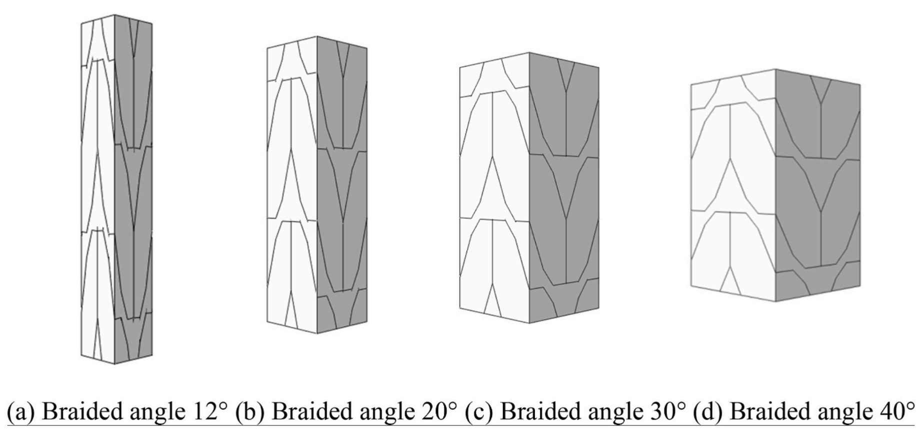

Fig. 8. Interior cell models with different braided angles.

and Tsai-Wu criterion is

$$\begin{aligned}&F_{11}\sigma_1^2+F_{22}\sigma_2^2+F_{33}\sigma_3^2+F_{44}\sigma_{23}^2+F_{55}\sigma_{13}^2+F_{66}\sigma_{12}^2\\&+2F_{12}\sigma_1\sigma_2+2F_{13}\sigma_1\sigma_3+2F_{23}\sigma_2\sigma_3+F_1\sigma_1+F_2\sigma_2+F_3\sigma_3=1\end{aligned} \tag{21}$$

in which

$$\begin{aligned}&F_{11}=\frac{1}{X_TX_C},F_{22}=F_{33}=\frac{1}{Y_TY_C},F_{44}=\frac{1}{S_{23}^2},F_{55}=\frac{1}{S_{13}^2},F_{66}=\frac{1}{S_{12}^2},\\&F_{12}=F_{13}=-\frac{1}{2}\sqrt{F_{11}F_{22}},F_{23}=-\frac{1}{2}\sqrt{F_{22}F_{33}},F_1=\frac{1}{X_T}-\frac{1}{X_C},F_2=F_3=\frac{1}{Y_T}-\frac{1}{Y_C}\end{aligned} \tag{22}$$

in the criteria above, X_T and X_C are the axial tensile and compression strength, Y_T, and Y_C are the transverse tensile and compression strength, and S_{12}, S_{13}, and S_{23} are the shear strength in three directions.

The initial damage of a fiber bundle is detected according to Hashin and Tsai Wu criteria, and the usdfld subroutine is utilized to simulate the stiffness reduction process on the Abaqus platform. Fig. 9 shows the stress-strain curves of 3-D 4-directional braided composites with a braided angle of 12°. It could be seen that the stress-strain behavior calculated by the Tsai-Wu criterion is in good agreement with the experimental results reported in reference (Liang et al., 2020). The strength obtained according to the Hashin criterion is smaller than that of the experimental data. This discrepancy could be attributed to the Hashin criterion overestimating the influence of shear damage, leading to premature failure of the composite. Hence, considering accuracy, the Tsai-Wu criterion is applied in this study to compute the strength of 3-D 4-directional braided composites with different braided angles, which are utilized to forecast the strength properties of braided composite plates with an eccentric hole.

4.2. Strength analysis of 3-D 4-directional braided composites with an eccentric hole

In this paper, the tensile strength of 3-D composite plates with a hole is predicted by theoretical calculation and simulation analysis. The theoretical methods include the average stress criterion (ASC) and the point stress criterion (PSC). Based on the ASC method, the ultimate

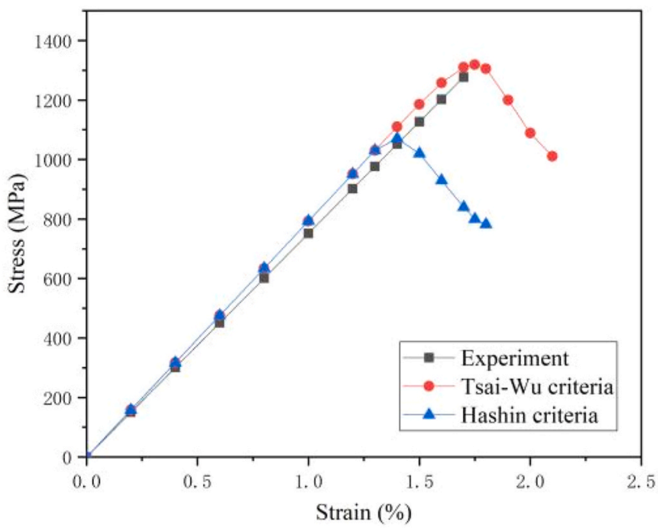

Fig. 9. Stress-strain curves of the composites with a braided angle of 12°.

failure of notched composites is due to the high-stress distribution in the area close to the hole, i.e. when the average stress over a finite distance reaches the un-notched strength of the laminate σ_0:

$$\frac{1}{a_0}\int_0^{a_0}\sigma_y(x,0)dx=\sigma_0 \tag{23}$$

where a_0 is the characteristic length, σ_0 is the tensile strength of the defect-free (unnotched) composite. The PSC method, on the other hand, indicates that the failure of the plate occurs when the stress level at a certain point near the hole edge (Whitney and Nuismer, 1974) reaches the un-notched strength of the composite:

$$\sigma_N=\frac{2\sigma_0}{2+\left(\frac{R}{R+d_0}\right)^2+3\left(\frac{R}{R+d_0}\right)^4-(K-3)\left(5\left(\frac{R}{R+d_0}\right)^6-7\left(\frac{R}{R+d_0}\right)^8\right)} \tag{24}$$

According to references (Whitney and Nuismer, 1974) and (Nuismer, 1978), the values of a_0 and d_0 of braided composites with different hole radius is shown in Table 9.

The simulations for the strength properties of 3-D 4-directional braided composites with an eccentric hole based on the Tsai-Wu, Hashin, and Hoffman criteria are carried out.

The notched strength of the braided composites with the braided angle of 12° and hole radii of 0.5 mm, 1.4 mm, 2.5 mm, 3.9 mm, and 4.8 mm are presented in Table 10. It is shown that the results attained from the PSC and ASC models and the Tsai-Wu criterion reveal good consistency with the experiments. However, the results calculated by the PSC and ASC methods are highly dependent on the values a_0 and d_0; it is difficult to predict the strength of the braided composites with an eccentric hole without any experimental data to determine the characteristic length. Moreover, the notched strength predictions by the PSC and ASC models are generally lower than the experimental measurements, possibly due to overestimating the stress concentration in the simulations and using damage criteria that aimed for monolayer structures. This observation further indicates that 3-D braided composites possess higher damage tolerance than laminates.

The strength of the 3-D 4-directional braided composites with an eccentric hole calculated according to the Tsai-Wu criterion is shown in Fig. 10. It can be seen that the strength of the composites decreases with the increase of the braided angles, hole radii, and eccentric distances. The eccentric distance δ shows little effect on the strength of the composite plate when the hole radius is 0.5 mm, where the stress distribution in such a plate is similar to that of an infinite plate. With the enlargement of the hole radius, the strength of the braided composites decrease significantly with the increase of the eccentric distance. It is due to the remarkable stress concentration that could be set off in a small ligament between the hole edge and the plate edge, which makes the composite plate with an eccentric hole more vulnerable under loading.

5. Conclusions

A muti-scale analysis method for the mechanical properties of 3-D 4-directional braided composites with an eccentric hole is proposed in this paper. Adopting the homogenization method, the macro equivalent stiffness of the braided composites is studied based on the configuration of the interior-, surface-, and corner cells. By revising the Lekhnitskii theory, an LSC (local stress correction) method for the orthotropic materials with an eccentric hole under tension was established. The simulation for 3-D 4-directional braided composites with an eccentric hole was then carried out, and the validity of the theoretical method with various parameters was verified.

The interior cell models of the composites with different braided angles were constructed by assuming an octagonal cross-section. The strength of defect-free braided composites was simulated according to different criteria using a user-defined subroutine in Abaqus. Compared to the existing experimental data, it was found that the Tsai-Wu criterion shows better accuracy than other commonly used failure criteria. The strength properties of 3-D 4-directional braided composites with a hole were then obtained by different theoretical methods and simulation analysis. The tensile strength of the composites with an eccentric hole with various parameters was predicted, which shows that the braided angle, hole radius, and eccentric distance affect the strength of the composites. With the enlargement of the hole, the tensile strength of the braided composites is significantly influenced by the eccentric distance.

The multi-scale analysis method proposed in this study can be utilized to estimate the stress distribution and the tensile strength of 3-D 4-

Table 9
Values of a_0 and d_0.

R(mm)	0.5	1.4	2.5	3.9	4.8
a_0(mm)	3.2	3.4	3.6	3.7	3.8
d_0(mm)	0.76	0.85	1.1	1.16	1.2

Table 10
Predictions for the strength of 3-D braided composite with a hole (MPa).

R(mm)	0.5	1.4	2.5	3.9	4.8
Experiment (Liang et al., 2021)	–	1100	970	786	674
PSC	1188.15	962.83	875.72	763.50	717.25
ASC	1151.46	979.29	861.54	763.37	718.07
Tsai-Wu	1183.23	1032.19	890.67	738.67	644.33
Hashin	1162.67	1020.16	878.67	718.80	626.00
Hoffman	1160.87	1013.33	876.00	713.41	621.19

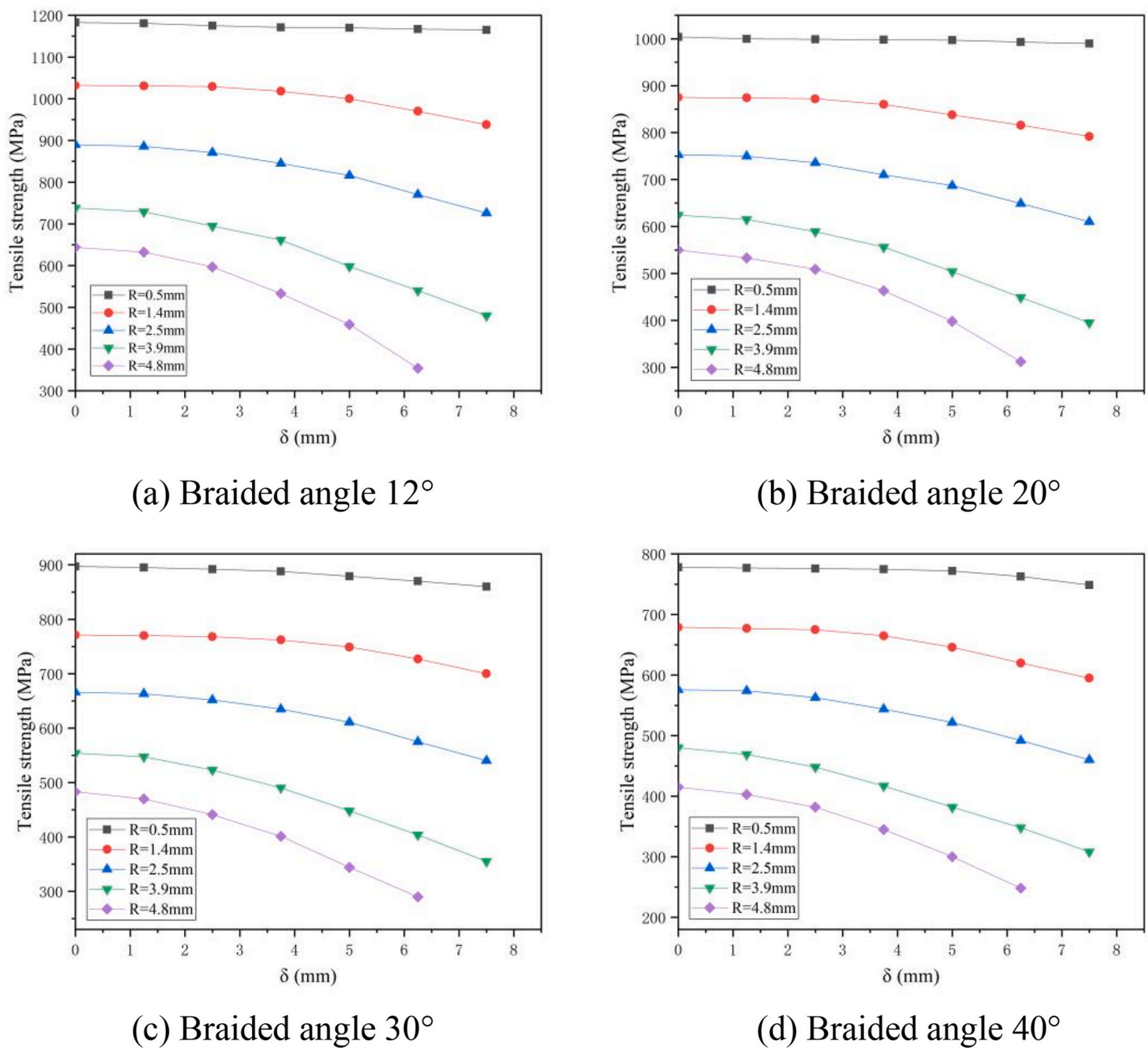

Fig. 10. Strength predictions of the braided composite with an eccentric hole according to the Tsai-Wu criterion.

directional braided composites with an eccentric hole. It provides theory and simulation methods for the problems of opening plates in engineering.

Author statement

Chentong Gao: Conceptualization, Data Curation, Investigation, Software, Formal Analysis, Visualization, Writing - Original Draft; Jianping Gu: Data Curation, Investigation, Formal Analysis; Hao Zeng: Investigation, Funding Acquisition; Akbar A. Khatibi (Corresponding Author 1): Supervision, Writing - Review & Editing; Huiyu Sun (Corresponding Author 2): Conceptualization, Supervision, Funding Acquisition, Writing - Review & Editing.

Declaration of competing interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Data availability

Data will be made available on request.

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